



GOSFORD HIGH SCHOOL

2008 HIGHER SCHOOL CERTIFICATE

MATHEMATICS ASSESSMENT TASK #1

DECEMBER 2007

Time Allowed – 70 minutes

All necessary working should be shown.

Full marks may not be awarded for unnecessarily untidy work or work that is poorly organized.

Students must begin each new question on a new page.

Students need to place their name and/or HSC candidate number at the top of each new page.

Questions will be collected separately at the conclusion of the assessment task.

All questions are to be attempted.

Question 1 (12 marks)

- (a) Find $\frac{d}{dx}[x^3 - 4x^2 + 12]$ (1)
- (b) Find $\lim_{x \rightarrow 0} \left[\frac{x}{x^2 - 2x} \right]$ (2)
- (c) Write a quadratic equation whose roots are $1 \pm \sqrt{3}$. (2)
- (d) A population (P) is increasing but at a decreasing rate.
Describe the signs of $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ where t is time. (2)
- (e) Find the equation of the locus of a point $P(x, y)$ which moves so that it is equidistant from the point $(0, 6)$ and the line $y = -6$. (1)
- (f) Find the primitive function of $4x^3 - 6x^2 + x$. (2)
- (g) It is given that a stationary point occurs at $x = 0$ on a continuous curve with $f''(x) = x^2(x - 2)(x - 4)$. Determine the nature of the stationary point at $x = 0$. (2)

Question 2 (12 marks)

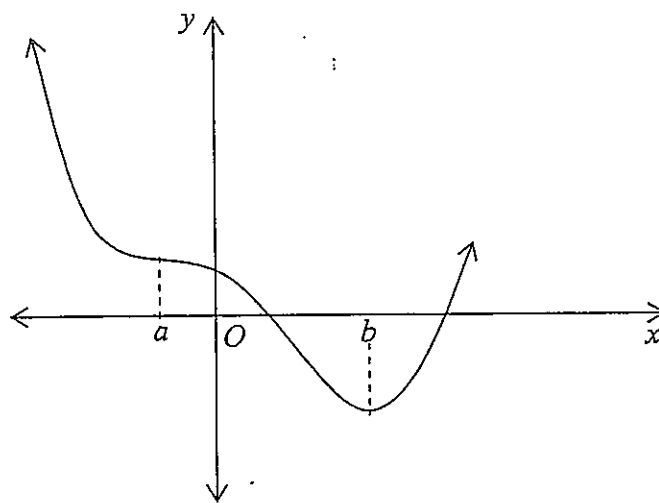
- (a) Find $f'(x)$ if $f(x) = x\sqrt{x} - \frac{3}{x^2}$ (3)
- (b) Find $\frac{dy}{dx}$ if $y = \frac{2x-1}{x^2+1}$ (3)
- (c) If α and β are the roots of the quadratic equation $2x^2 - 6x + 7$
- (i) find the value of $\alpha + \beta$ (1)
- (ii) find the value of $\alpha\beta$ (1)
- (iii) find the value of $(\alpha - \beta)^2$ (2)
- (iv) Are the roots of the equation Real or Unreal? Explain your answer. (2)

Question 3 (12 marks)

- Find the coordinates of the focus of the parabola with equation $(x - 4)^2 = 16(y + 6)$ (2)
- Find the centre and radius of a circle with equation $x^2 + y^2 - 8x = 0$. (2)
- Write down the equation of the parabola with axis the x axis and vertex the origin and passing through the point $(-2, 6)$. (2)
- Find the constants k and g such that $x^2 + 10x + 10 \equiv k(x + 2)^2 + g(x + 1)$. (3)
- For what values of m are the roots of the equation $x^2 + 2mx + 2(m + 12) = 0$ real? (3)

Question 4 (12 marks)

- The graph below represents the function $y = f(x)$.
Using the provided enlarged copy of this diagram, on the same set of axes graph the gradient function $y = f'(x)$ (2)



- Find the equation of the curve with gradient function $(1 + 2x)^3$ if the curve passes through the point $(\frac{1}{2}, 0)$. (3)
- Find the equation of the tangent to the parabola $y = x^2 - 3x - 4$ at the point on the parabola where the tangent is parallel to the line $y = 2 - x$. (4)
- A and B are the points $(-3, 0)$ and $(3, 0)$ respectively.
Find the equation of the locus of the point $P(x, y)$ which moves such $PA = 2PB$ (3)

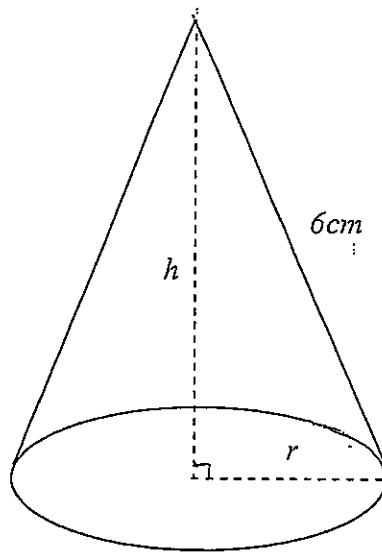
Question 5 (12 marks)

- (a) (i) Find the stationary points on the curve $y = 2x^3 - 3x^2 - 12x$ and determine their nature. (4)
- (ii) Sketch the curve in the domain $-2 \leq x \leq 3$ indicating on your sketch the x and y intercepts and all critical points. (do not attempt to find any points of inflexion) (3)

- (b) The slant edge of a right circular cone is 6cm in length.

The volume(V) of a cone is given by the formula $V = \frac{1}{3}\pi r^2 h$

- (i) Show that for the given cone below $V = 12\pi h - \frac{\pi h^3}{3}$. (2)
- (i) Hence, or otherwise, find the height of the cone when the volume is a maximum. (3)



1/

a) $\frac{d}{dx} (x^3 - 4x^2 + 12) = 3x^2 - 8x$

b) $\lim_{x \rightarrow 0} \left[\frac{x}{x^2 - 2x} \right] = \lim_{x \rightarrow 0} \left[\frac{1}{x - 2} \right]$
 $= -\frac{1}{2}$

c) $\alpha = 1 + \sqrt{3} \quad \beta = 1 - \sqrt{3}$

$\alpha + \beta = 2 \quad \alpha\beta = -2$

$x^2 - 2x - 2 = 0$

d) $\frac{dP}{dt} > 0 \quad \frac{d^2P}{dt^2} < 0$

e) $a = 6$

$x^2 = 24y$

f) $\int (4x^3 - 6x^2 + x) dx = x^4 - 2x^3 + \frac{x^2}{2} + c$

g) $f''(x) = x^2(x-2)(x-4)$

Test	x	$f''(x)$
	0^-	> 0
	0^+	> 0

$\therefore$ No change in concavity

$\therefore$ Min. St pt at $x = 0$.

Q2 a) $f(x) = x\sqrt{x} - \frac{3}{x^2}$

$= x^{3/2} - 3x^{-2}$

$f'(x) = \frac{3}{2}x^{1/2} + 6x^{-3}$

b) $y = \frac{2x-1}{x^2+1}$

$\frac{dy}{dx} = \frac{(x^2+1)^2 - (2x-1)2x}{(x^2+1)^2}$
 $= \frac{-2x^2 + 2x + 2}{(x^2+1)^2}$

c) $2x^2 - 6x + 7$

(i) $\alpha + \beta = 3$

(ii) $\alpha\beta = 3\frac{1}{2}$

iii) $(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$
 $= (\alpha + \beta)^2 - 4\alpha\beta$
 $= 9 - 14$
 $= -5$

iv) Roots un-real since $(\alpha - \beta)^2 < 0$

Q3

a) $(y-4)^2 = 16(x+6)$
 $V(4, -6) \quad a = 4$

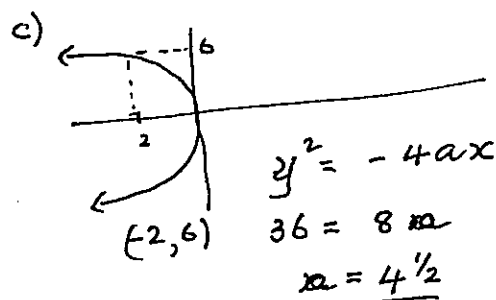
$F(4, -2)$

b) $x^2 + y^2 - 8x = 0$

$(x-4)^2 + y^2 = 16$

Centre $(4, 0)$

radius = 4.



$\therefore y^2 = -18x$

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x) (i) $y = 2x^3 - 3x^2 - 12x$
 $y' = 6x^2 - 6x$
 $= 6x(x-1)$
 $= 0$ at $(0,0)$ $(1, -14)$

$y'' = 12x - 6$
 $= -6$ at $(0,0)$
 $= 6$ at $(1, -14)$

$\therefore$ Min stpt $(1, -14)$

Max stpt $(0,0)$

when $x = -2$

$y = -16 - 12 + 24$
 $= -4$

when $x = 3$ $y = 54 - 27 - 36$
 $= -9$

Inflexion pts occur when $y'' = 0$
 $\therefore x = \frac{1}{2}$ $y = -6\frac{1}{2}$

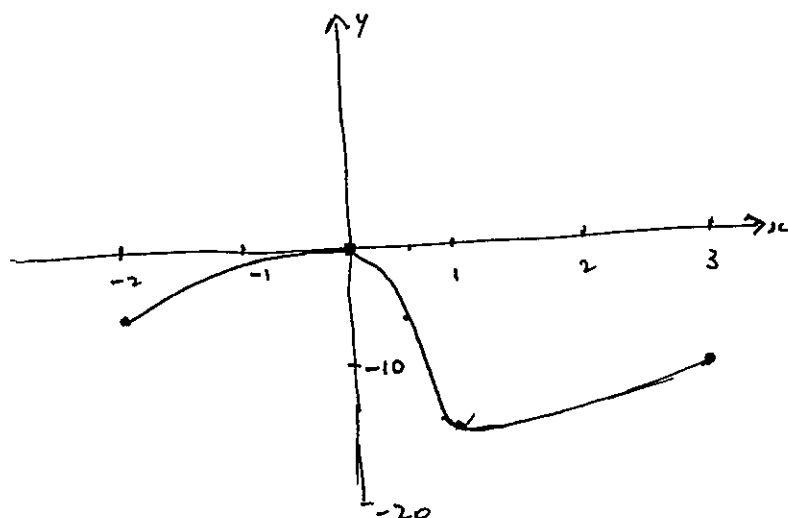
Test:

x	y''
$\frac{1}{2}^-$	< 0
$\frac{1}{2}^+$	> 0

$\therefore$ change in concavity
at $(\frac{1}{2}, -6\frac{1}{2})$

$\therefore$ Inflexion pt.

ii)



b) $h^2 + r^2 = 36$
 $r = \sqrt{36 - h^2}$

(i) $V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi (36 - h^2) h$
 $= 12\pi h - \frac{\pi h^3}{3}$

$\frac{dV}{dh} = 12\pi - \pi h^2$
 $= 0$ when $h = \sqrt{12}$

$\frac{d^2V}{dh^2} = -2\pi h$
 < 0 when $h = \sqrt{12}$

$\therefore$ Max V when $h = \sqrt{12}$ cm